

```

Dataset Shape: (581012, 54)
Target Shape: (581012,)
Classes: [1 2 3 4 5 6 7]

Class Distribution:
1    211840
2    283301
3    35754
4     2747
5     9493
6    17367
7    20510
Name: count, dtype: int64
Imbalance Ratio: 103.13105205678923
One-Hot Target Shape: (581012, 7)

Training: (464809, 54)
Validation: (58101, 54)
Testing: (58102, 54)

Class Weights:
Class 0: Count=169472, Weight=0.3918
Class 1: Count=226640, Weight=0.2930
Class 2: Count=28603, Weight=2.3215
Class 3: Count=2198, Weight=30.2099
Class 4: Count=7594, Weight=8.7439
Class 5: Count=13894, Weight=4.7791
Class 6: Count=16408, Weight=4.0469

Network Architecture: 54 -> 64 -> 32 -> 16 -> 7

```

Starting Training...

Epoch	1		Train Loss: 0.6900		Val Loss: 0.7041		Train Acc: 0.5896		Val Acc: 0.5897
Epoch	2		Train Loss: 0.6319		Val Loss: 0.6388		Train Acc: 0.6174		Val Acc: 0.6167
Epoch	3		Train Loss: 0.5771		Val Loss: 0.5888		Train Acc: 0.6476		Val Acc: 0.6473
Epoch	4		Train Loss: 0.5602		Val Loss: 0.5766		Train Acc: 0.6840		Val Acc: 0.6859
Epoch	5		Train Loss: 0.5388		Val Loss: 0.5516		Train Acc: 0.6751		Val Acc: 0.6759
Epoch	6		Train Loss: 0.5109		Val Loss: 0.5191		Train Acc: 0.6814		Val Acc: 0.6834
Epoch	7		Train Loss: 0.5210		Val Loss: 0.5357		Train Acc: 0.7005		Val Acc: 0.7023
Epoch	8		Train Loss: 0.4897		Val Loss: 0.4995		Train Acc: 0.6698		Val Acc: 0.6701
Epoch	9		Train Loss: 0.4800		Val Loss: 0.4872		Train Acc: 0.7132		Val Acc: 0.7148
Epoch	10		Train Loss: 0.4697		Val Loss: 0.4797		Train Acc: 0.7105		Val Acc: 0.7125
Epoch	11		Train Loss: 0.4549		Val Loss: 0.4629		Train Acc: 0.7086		Val Acc: 0.7101
Epoch	12		Train Loss: 0.4971		Val Loss: 0.5088		Train Acc: 0.6866		Val Acc: 0.6897
Epoch	13		Train Loss: 0.4374		Val Loss: 0.4490		Train Acc: 0.7142		Val Acc: 0.7161
Epoch	14		Train Loss: 0.4411		Val Loss: 0.4533		Train Acc: 0.7141		Val Acc: 0.7156
Epoch	15		Train Loss: 0.4310		Val Loss: 0.4470		Train Acc: 0.7078		Val Acc: 0.7072
Epoch	16		Train Loss: 0.4205		Val Loss: 0.4311		Train Acc: 0.7450		Val Acc: 0.7445
Epoch	17		Train Loss: 0.4178		Val Loss: 0.4290		Train Acc: 0.7269		Val Acc: 0.7280
Epoch	18		Train Loss: 0.4082		Val Loss: 0.4204		Train Acc: 0.7296		Val Acc: 0.7289
Epoch	19		Train Loss: 0.4040		Val Loss: 0.4166		Train Acc: 0.7481		Val Acc: 0.7489
Epoch	20		Train Loss: 0.4026		Val Loss: 0.4239		Train Acc: 0.7337		Val Acc: 0.7349

.....Till 100 epochs.

FINAL TEST RESULTS

Test Accuracy: 81.12%
Macro Precision: 0.6839
Macro Recall: 0.8879
Macro F1 Score: 0.7510

Classification Report:

	precision	recall	f1-score	support
Spruce/Fir	0.8442	0.7918	0.8172	21184
Lodgepole Pine	0.8594	0.8023	0.8299	28331
Ponderosa Pine	0.8288	0.8098	0.8192	3576
Cottonwood/Willow	0.5460	0.9745	0.6999	274
Aspen	0.3738	0.9737	0.5402	949
Douglas-fir	0.5644	0.8935	0.6918	1737
Krummholz	0.7706	0.9698	0.8588	2051
accuracy			0.8112	58102
macro avg	0.6839	0.8879	0.7510	58102
weighted avg	0.8306	0.8112	0.8161	58102

Confusion Matrix:

```
[[16773 3678 7 0 221 24 481]
 [ 3035 22730 463 6 1277 709 111]
 [ 0 22 2896 164 40 454 0]
 [ 0 0 3 267 0 4 0]
 [ 0 15 3 0 924 7 0]
 [ 0 1 122 52 10 1552 0]
 [ 60 2 0 0 0 0 1989]]
```

Per-Class Metrics:

Class	Precision	Recall	F1 Score
Spruce/Fir	0.844222	0.791777	0.817159
Lodgepole Pine	0.859422	0.802301	0.829880
Ponderosa Pine	0.828849	0.809843	0.819236
Cottonwood/Willow	0.546012	0.974453	0.699869
Aspen	0.373786	0.973656	0.540193
Douglas-fir	0.564364	0.893495	0.691776
Krummholz	0.770632	0.969771	0.858808

MODEL SUMMARY

Architecture: 54 -> 64 -> 32 -> 16 -> 7
Hidden Activation: ReLU
Output Activation: Softmax
Loss: Weighted Categorical Cross-Entropy
Optimizer: Mini-Batch SGD
Batch Size: 128
Learning Rate: 0.01
Test Accuracy: 81.12%
Macro F1 Score: 0.7510





